ShortcutBuster

Detecting and fixing "shortcut learning" in image classification models

Project Documentation - Part 1
Planning & Management | Literature Review | Requirements Gathering

Digital Egypt Pioneers Initiative (DEPI) - Graduation Project

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Technical Instructor: Mostafa Asala

GitHub: https://github.com/zahraelshafey/ShortcutBuster

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Note: items marked "proposed" are the team's current plan and may be refined with the Technical Instructor as the project progresses.

1. Project Planning & Management

1.1 Project Proposal

Overview. ShortcutBuster is a tool that detects when an image-classification model reaches high accuracy for the wrong reasons ("shortcut learning"), measures how much it depends on that shortcut, and retrains the model so it learns the real concept.

Problem. Models often learn easy shortcuts instead of the true feature. For example, a classifier that separates waterbirds from landbirds may simply look at the background (water or land). It looks accurate on typical data but fails when a waterbird appears on land. Geirhos et al. describe this as confusing performance on a dataset with real ability [1].

Objectives.

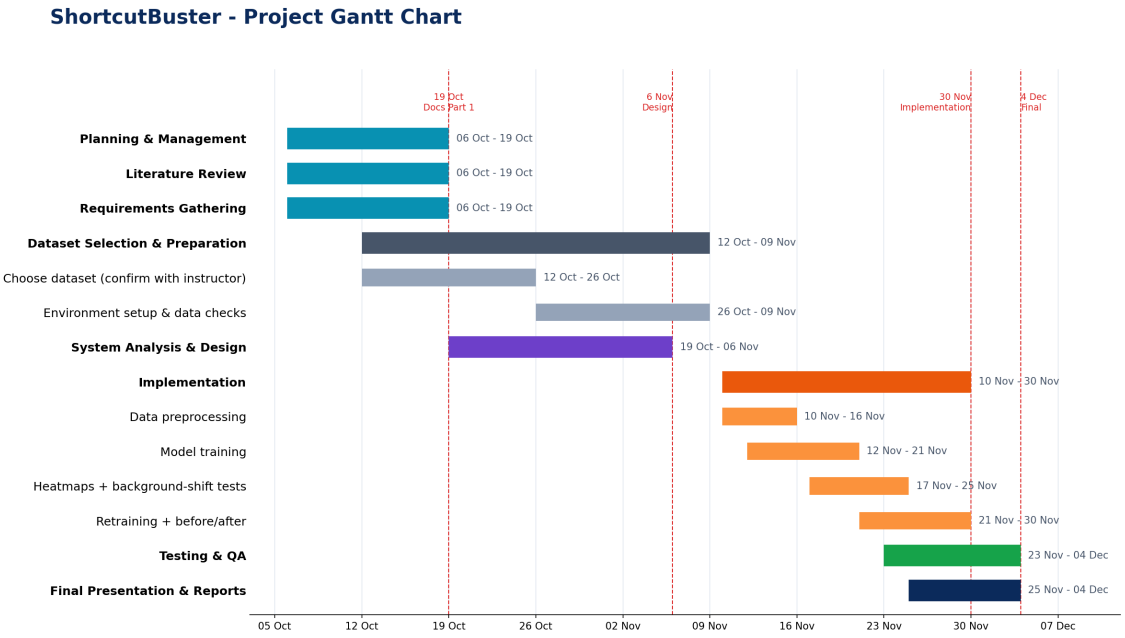
- Detect whether a trained model relies on a background shortcut.
- Identify what the model depends on, using visual explanations (heatmaps).
- Measure the accuracy drop when the shortcut stops working (per-group and worst-group accuracy).
- Retrain the model to reduce shortcut reliance and compare results before and after.
- Document and publish everything on GitHub in a reproducible way.

Scope.

- **In scope:** image classification on the Waterbirds benchmark (dataset choice is provisional, see Section 2.4); a baseline model; heatmap analysis; a background-shift test; retraining; a before/after evaluation report.
- **Out of scope:** other data types (text, audio), production-scale deployment, and detecting every possible kind of bias. The project focuses on one clear and measurable shortcut: the image background.

1.2 Project Plan

The plan below follows the phases and deadlines in the DEPI Project Documentation Guidelines. October is used to choose and prepare the dataset and to complete the documentation and design work; hands-on implementation starts on 10 November.



Milestones and deliverables

Milestone	Deliverables	Date
Documentation Part 1	Planning & Management, Literature Review, Requirements Gathering (one PDF on GitHub)	19 Oct 2026 (official announcement)
System Analysis & Design	Use case, architecture, ERD/schema, DFD, sequence, activity, state, class, deployment and component diagrams, wireframes	6 Nov 2026
Implementation	Source code, GitHub workflow, README with run instructions, executable/demo	30 Nov 2026
Final Presentation, Testing & Reports	Test plan and bug reports, user manual, technical documentation, presentation, infographic, optional demo video	4 Dec 2026

Resource allocation (proposed). A team of four; a free cloud notebook environment with GPU for training; a public GitHub repository for code, documents and issue tracking; the Waterbirds benchmark dataset; and weekly review meetings with the Technical Instructor.

1.3 Task Assignment & Roles

Proposed distribution of responsibilities (to be confirmed by the team):

Member	Role	Responsibilities
Zahra Ahmed Hamada	Team Leader, Model Training & Documentation	Planning and follow-up, GitHub repository and documentation, model training (with Sondos), coordination with the Technical Instructor
Rana Haitham	Data Preprocessing & Evaluation Metrics	Dataset preparation and preprocessing (with Menna-Allah), then evaluation metrics: per-group and worst-group accuracy
Sondos Abdo	Model Training & Explainability	Model training (with Zahra), heatmaps (visual explanations), background-shift tests
Menna-Allah Gharib	Data Preprocessing & Testing	Dataset preparation and preprocessing (with Rana), then before/after comparison, test cases and bug reports

Member	Role	Responsibilities
Technical Instructor: Mostafa Asala	Supervision	Reviews, feedback and approval of project direction

1.4 Risk Assessment & Mitigation Plan

Risk	Likelihood	Impact	Mitigation
Limited GPU / compute power	Medium	High	Use small pretrained models and free cloud notebooks; start with a small baseline early
Downloaded dataset copy lacks group metadata (bird type and background)	Medium	High	Verify the metadata file first; fall back to the official source if needed
Retraining reduces overall accuracy	Medium	Medium	Track both average and worst-group accuracy; tune the mitigation; agree an acceptable margin with the instructor
The fix does not fully remove the shortcut (related work still reports remaining errors [3])	High	Medium	Report results honestly with clear metrics; iterate on the method; list limits in the final report
Schedule slips on tight deadlines	Medium	High	Weekly reviews, early baseline, tasks tracked as GitHub issues
Team member unavailable	Low	Medium	Clear roles, shared repository, a backup owner for each critical task
Results are not reproducible	Low	High	Fixed random seeds, versioned configuration, README with exact run steps

1.5 Key Performance Indicators (KPIs)

Targets are defined relative to the baseline model and will be fixed numerically once the baseline results exist.

KPI	Definition	Success criterion
Worst-group accuracy	Accuracy on the weakest of the four groups (bird type x background)	Increases after retraining
Accuracy gap	Average accuracy minus worst-group accuracy	Decreases after retraining
Shortcut reliance	Change in predictions when the background shortcut is removed or swapped (inspired by the MAD and flip-rate metrics in [3])	Decreases after retraining
Overall accuracy retained	Average accuracy before vs. after retraining	Not significantly lower (margin agreed with instructor)
Reproducibility	Experiments can be rerun from the README	All reported results reproducible
Delivery on time	Documents and code delivered by the official deadlines	100% of milestones met

2. Literature Review

This section reviews similar projects and solutions for detecting and reducing shortcut learning, summarises their strengths and weaknesses, compares them with ShortcutBuster, and documents the sources used.

2.1 Background

Shortcut learning means that a model relies on easy, unintended cues that work on the training data but fail in real conditions [1]. The Waterbirds benchmark was built to study this: bird photos are combined with water or land backgrounds so that, in training, the background strongly predicts the label. Models are then judged by their worst-group accuracy, not only their average accuracy [2].

2.2 Similar Projects and Solutions

Work	What it does	Strengths	Limitations
Nauta et al., 2022 [3]: shortcut learning in skin cancer models	Adds or removes a known artefact (coloured patches) using image inpainting, and measures how predictions change in an already trained model. Then retrains on a de-biased dataset.	Detects, quantifies and fixes in one workflow; model-agnostic detection; clear metrics (MAD, flip rate); code on GitHub. Retraining cut the flip rate on benign images from 21.1% to 0.6%.	Needs the artefact to be easy to identify and segment; needs an inpainting model; medical images only; not a reusable tool; the fix was incomplete (33.5% of malignant images still flipped).
Group DRO, Sagawa et al., 2020 [2]	Trains the model to minimise the worst-group error, using known group labels.	Improves worst-group accuracy; defines the Waterbirds benchmark used by many papers.	Needs group labels during training; does not by itself tell the user whether a given model is cheating.
Just Train Twice, Liu et al., 2021 [4]	Trains a model, finds the examples it gets wrong, and trains again giving them more weight.	No group labels needed in training; simple two-stage idea.	Depends on hyperparameters; can amplify unwanted samples (see [6]).
Grad-CAM, Selvaraju et al., 2017 [5]	Produces heatmaps showing which image regions influenced a prediction.	Simple, widely used, easy to interpret visually.	Qualitative; shows where the model looked but does not measure reliance or fix it.
Panaitescu-Liess et al., 2024 [6]	Shows that group-robustness methods without group labels can also amplify poisoned samples, and that poisoning defences can hurt minority groups.	Explains important risks and trade-offs of fixing methods.	Analysis paper, not a tool.
EyeBirds, Zhou et al., 2022 [7]	A smartphone app recognising 548 water bird species using an attention-based CNN; bird masks emphasise the bird region over the background.	Complete app and dataset; shows that background can hurt bird classification.	Targets species recognition and accuracy; does not measure or diagnose reliance on the background.

2.3 Comparison with ShortcutBuster

Existing work either explains models (Grad-CAM), improves worst-group performance with a training method (Group DRO, Just Train Twice), or studies shortcuts in a different domain (Nauta et al.).

ShortcutBuster combines the full workflow in one project for bird images:

- **Detect** the shortcut with per-group accuracy and heatmaps.
- **Measure** reliance with the worst-group accuracy and shift-based metrics adapted from [3].
- **Fix** by retraining with a chosen mitigation (final method to be agreed with the instructor), then report **before vs. after**.

Lessons we apply: use quantitative metrics rather than only heatmaps [3,5]; always report worst-group and average accuracy together [2]; be transparent that fixes may be partial [3] and may carry risks [6].

2.4 Dataset Choice (provisional)

The team reviewed three candidate datasets on Kaggle:

Dataset	Type of bias studied	Assessment
Waterbirds	Background bias; four groups (bird type x background); worst-group accuracy applies directly	**Provisional choice.** Clear, standard benchmark with published baselines, easy to explain.
CelebA	Attribute and gender bias in face images	Possible second dataset if time allows; larger and more sensitive in subject.
PACS	Style and domain shift (photo, art, cartoon, sketch)	Measures domain generalisation rather than a labelled shortcut; kept as future work.

The final dataset choice will be confirmed with the Technical Instructor before implementation begins.

2.5 References

[1] Geirhos, R. et al. "Shortcut learning in deep neural networks." *Nature Machine Intelligence* 2, 665-673 (2020).

[2] Sagawa, S., Koh, P. W., Hashimoto, T., Liang, P. "Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization." *ICLR* 2020.

[3] Nauta, M., Walsh, R., Dubowski, A., Seifert, C. "Uncovering and Correcting Shortcut Learning in Machine Learning Models for Skin Cancer Diagnosis." *Diagnostics* 12(1), 40 (2022). https://doi.org/10.3390/diagnostics12010040

[4] Liu, E. Z. et al. "Just Train Twice: Improving Group Robustness without Training Group Information." *ICML* 2021.

[5] Selvaraju, R. R. et al. "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization." *ICCV* 2017.

[6] Panaitescu-Liess, M.-A., Kaya, Y., Zhu, S., Huang, F., Dumitras, T. "Like Oil and Water: Group Robustness Methods and Poisoning Defenses May Be at Odds." *ICLR* 2024. https://proceedings.iclr.cc/paper_files/paper/2024/hash/ff7c9c90030f1eb3cbf5c81b6fbd9a05-Abstract-Conference.html

[7] Zhou, J. et al. "Eyebirds: Enabling the Public to Recognize Water Birds at Hand." *Animals* 12(21), 3000 (2022). https://doi.org/10.3390/ani12213000

[8] Waterbirds dataset (Kaggle): https://www.kaggle.com/datasets/satwiksps/waterbirds-spurious-correlation-benchmark

[9] CelebA dataset (Kaggle): https://www.kaggle.com/datasets/jessicali9530/celeba-dataset

[10] PACS dataset (Kaggle): https://www.kaggle.com/datasets/praveen1441/pacs-dataset

3. Requirements Gathering

3.1 Stakeholder Analysis

Stakeholder	Description	Needs
ML students and engineers (primary users)	People who train image classifiers	Check whether their model cheats, understand why, and fix it with clear evidence
Researchers and educators	People studying bias and robustness	Reproducible experiments on a standard benchmark, with clear metrics
Technical Instructor and DEPI evaluators	Supervise and assess the project	Complete documentation, working code, tests and a clear demonstration
Development team	Zahra, Rana, Sondos, Menna-Allah	Clear scope, modular design and shared tracking of tasks
Organisations relying on AI (indirect)	Teams deploying image models	Models they can trust beyond benchmark accuracy

3.2 User Stories & Use Cases

ID	User story
US-1	As a user, I want to load a dataset and a model so I can analyse it.
US-2	As a user, I want to see accuracy for each group so I can tell whether the model cheats.
US-3	As a user, I want heatmaps so I can see where the model looks in the image.
US-4	As a user, I want to test the model on shifted backgrounds so I can measure the accuracy drop.
US-5	As a user, I want to retrain the model and compare before and after results.
US-6	As an evaluator, I want to rerun the experiments from the README to verify the results.

Use case	Actor	Main flow
UC-1 Detect shortcut reliance	User	Load data and model; compute per-group accuracy; generate heatmaps; view the report
UC-2 Measure impact of the shortcut	User	Run the background-shift test; compute worst-group accuracy and the accuracy gap
UC-3 Retrain and compare	User	Select the mitigation; retrain; re-evaluate; view before/after charts and metrics

3.3 Functional Requirements

ID	Requirement
FR-1	Load the dataset together with its metadata (class and group labels).
FR-2	Train a baseline image classification model.
FR-3	Compute accuracy per group, the average accuracy and the worst-group accuracy.
FR-4	Generate heatmaps showing which image regions influence predictions.
FR-5	Run a background-shift test and compute shortcut-reliance metrics.
FR-6	Retrain the model with a selected mitigation method.
FR-7	Produce a before/after comparison report with tables and charts.
FR-8	Save configurations, results and trained models so experiments can be repeated.
FR-9	Provide a simple way to run the workflow and view results (interface to be defined in the design phase).

3.4 Non-functional Requirements

Category	Requirement
Performance	Training and evaluation must run on a single GPU or a free cloud notebook within a reasonable time.
Reliability and reproducibility	Fixed random seeds and versioned configuration so results can be reproduced.
Usability	Clear visual outputs and a README that lets a new user run the project step by step.
Security and compliance	No personal data is processed; dataset licences are respected; dependencies are listed and pinned.
Maintainability	Modular, well-commented code organised in separate components; clear naming and documentation.
Portability	Documented environment setup that works on common operating systems and cloud notebooks.